

HERNIA

Definition & composition

- **Hernia** = protrusion of a viscus (or part) through an abnormal opening in the wall of its containing cavity.
- **Sac** = peritoneal diverticulum (mouth, neck, body, fundus).
- **Coverings** = layers of abdominal wall it passes through.
- **Contents**: omentum (omenterocele), intestine (enterocele), part of bowel circumference (**Richter's**), bladder, ovary, Meckel's diverticulum (**Littre's**).

Aetiology & risk factors

- Sites of weakness. **Congenital** or **acquired**.
- Risk = ↑intra-abdominal pressure: heavy lifting, chronic cough, constipation, straining (BPH), pregnancy, obesity.

Reducibility (key terms)

- **Reducible**: returns spontaneously/manually.
- **Irreducible** subtypes:
- **Incarcerated**: adhesions sac↔contents; NO obstruction, NO vascular compromise.
- **Obstructed**: hollow viscus trapped → obstruction; blood supply intact (common cause of SBO).
- **Strangulated**: arterial supply compromised → gangrene unless urgent surgery.

Classification

- **External**: inguinal (direct/indirect), femoral, umbilical, paraumbilical, epigastric, incisional; rare (spigelian, obturator, lumbar, gluteal).
- **Internal**: diaphragmatic, hiatus (sliding & paraoesophageal).

Inguinal canal anatomy

- **Anterior wall**: external oblique aponeurosis (+ internal oblique laterally).
- **Posterior wall**: transversalis fascia (+ conjoint tendon medially).
- **Roof**: internal oblique + transversus abdominis. **Floor**: inguinal ligament (+ lacunar ligament medially).
- Contents (male): spermatic cord (vas deferens, testicular/cremasteric/vas arteries, pampiniform plexus), ilioinguinal nerve, genital branch genitofemoral. (Female): round ligament of uterus.

Inguinal hernia — direct vs indirect

- **Indirect**: through **deep ring**, **LATERAL** to inferior epigastric vessels; patent processus vaginalis (congenital); sac can contain bowel.
- **Direct**: through **Hesselbach's triangle**, **MEDIAL** to inferior epigastric vessels; weakness of posterior wall (acquired); usually retroperitoneal fat.
- **Pantaloon hernia** = both direct + indirect straddling inferior epigastric artery.
- **Hesselbach's triangle**: medial = lateral border rectus abdominis; lateral = inferior epigastric vessels; inferior = inguinal ligament.

Inguinal hernia — clinical

- History: age, occupation, swelling/discomfort/pain, systemic (if obstructed/strangulated: pain, vomiting, constipation, distension), precipitating factors.
- Exam: inspect (site/size/shape), palpate (reducibility, composition), **expansile cough impulse**, general exam for ↑intra-abdo pressure causes.
- **DDx (male)**: femoral hernia, vaginal hydrocele, spermatocele, inguinal lymphadenopathy, encysted hydrocele of cord, undescended testis, lipoma of cord, saphena varix, femoral aneurysm, psoas abscess, varicocele.
- **DDx (female)**: hydrocele of canal of Nuck (smooth, fluctuant, transilluminates, no cough impulse).

Femoral hernia

- Through **femoral canal**, presents in groin **below and lateral to pubic tubercle**. **More common in females**.
- Femoral sheath (lateral→medial): femoral artery, vein, canal. Canal contains **node of Cloquet**.
- **Femoral ring**: anterior = inguinal ligament; posterior = pectineal/iliopsoas; medial = lacunar ligament; lateral = femoral vein.
- **High risk of strangulation** → **repair soon after diagnosis**. **No truss**.
- Open approaches: **Lockwood (infra-inguinal)**, **McEvedy (supra-inguinal)**, **Lotheissen (trans-inguinal)**.

Inguinal hernia — management

- **Reducible** → surgery (mostly). **Irreducible** → emergency (strangulation risk); attempt reduction, **BUT if strangulation suspected do NOT reduce**.
- **Indirect** → **herniotomy** (dissect sac off cord, ligate at deep ring). Children = herniotomy only.
- **Direct** → **not opened, inverted** + herniorrhaphy.
- **Herniorrhaphy (floor reconstruction, all adult repairs)**:
- Primary tissue: **Bassini** (transversalis/conjoint → inguinal ligament), **McVay** (→ Cooper's ligament), **Shouldice** (incise + reapproximate).
- **Tension-free mesh**: **Lichtenstein**, plug & patch.
- Laparoscopic: **TAPP**, **TEP**.

Other hernias

- **Umbilical**: congenital (most close by 1 yr; repair if persists >3 yrs or defect >1.5cm); acquired (↑intra-abdo pressure — pregnancy, ascites, ovarian cyst, fibroid).
- **Paraumbilical**: above/below umbilicus; obese females; narrow neck → **high strangulation risk**; commonest content omentum. **Mayo repair** (overlapping); mesh if >4cm/recurrent.
- **Epigastric**: through linea alba, extraperitoneal fat; may trap/ischaemia. Repair simple suture; mesh if >4cm.
- **Incisional**: through previous scar; risk = age, obesity, wound infection/haematoma, ↑intra-abdo pressure, steroids, **midline > transverse incision**, poor technique. Wide neck → low strangulation risk. Mesh for large.
- **Richter's**: part of bowel wall (antimesenteric border) trapped; **lumen intact, NO obstruction** — but can strangulate.
- **Diaphragmatic**: traumatic (left > right, stomach/spleen herniate); hiatus — **sliding** (GOJ moves up) vs **paraesophageal** (GOJ fixed, stomach rotates → dangerous obstruction).

High-yield one-liners

- **Indirect = lateral to inferior epigastric (congenital); Direct = medial (acquired).**
- **Strangulated = compromised arterial supply → don't reduce, urgent surgery.**
- **Femoral hernia: below + lateral to pubic tubercle, females, high strangulation risk, repair early.**
- **Inguinal hernia: above + medial to pubic tubercle.**
- **Children inguinal hernia = herniotomy only; adults need floor repair (mesh/Lichtenstein).**
- **Paraumbilical = Mayo repair; narrow neck = strangulation risk.**
- **Richter's = partial bowel wall, lumen intact, no obstruction.**

HERNIA — Clinical Revision Table

Bold = high-yield. Hernia = protrusion of viscus through abnormal opening in its containing cavity wall.

DEFINITIONS & REDUCIBILITY

Term	Meaning
Sac / coverings / contents	Sac = peritoneal diverticulum; contents = omentum (omentocoele), bowel (enterocoele), Richter's (part of wall), Littre's (Meckel's)
Reducible	Returns spontaneously/manually
Incarcerated	Adhesions sac↔contents; NO obstruction, NO vascular compromise
Obstructed	Hollow viscus trapped → obstruction; blood supply intact (cause of SBO)
Strangulated	Arterial supply compromised → gangrene → urgent surgery ; DO NOT reduce

INGUINAL HERNIA — DIRECT vs INDIRECT

Feature	Indirect	Direct
Route	Through deep (internal) ring	Through Hesselbach's triangle
Vs inferior epigastric vessels	LATERAL	MEDIAL
Cause	Patent processus vaginalis (congenital)	Posterior wall weakness (acquired)
Contents	Sac ± bowel	Retroperitoneal fat (less often bowel)
Note	Pantaloon = both, straddle inferior epigastric a. Hesselbach: rectus (medial), inf. epigastric (lateral), inguinal lig (inferior)	

INGUINAL vs FEMORAL HERNIA

	Inguinal	Femoral
Position vs pubic tubercle	Above + medial	Below + latera l
Sex	M > F	F > M
Strangulation	Lower	High → repair early, NO truss
Ring boundaries	Inguinal canal	Femoral ring: ant=inguinal lig, post=pectineal, medial=lacunar lig, lateral=femoral vein
Approaches	Herniotomy ± herniorrhaphy	Open: Lockwood / McEvedy / Lotheissen

REPAIR

Type	Method
Indirect	Herniotomy (ligate sac at deep ring); children = herniotomy only
Direct	Inverted (not opened) + floor repair
Primary tissue	Bassini (→inguinal lig), McVay (→Cooper's), Shouldice
Tension-free mesh	Lichtenstein , plug & patch
Laparoscopic	TAPP, TEP

OTHER HERNIAS

Hernia	Key points
Umbilical	Congenital (close by 1yr; repair if >3yr or defect >1.5cm); acquired = ↑intra-abdo pressure
Paraumbilical	Obese females; narrow neck → strangulation risk ; Mayo repair ; mesh if >4cm
Epigastric	Linea alba, extraperitoneal fat; may trap/ischaemia; mesh if >4cm
Incisional	Previous scar; midline > transverse ; wide neck = low strangulation; mesh for large
Richter's	Part of bowel wall (antimesenteric); lumen intact, no obstruction , can strangulate

Hernia	Key points
Diaphragmatic / hiatus	Traumatic (L>R). Hiatus: sliding (GOJ up) vs paraoesophageal (GOJ fixed, can obstruct)